

Multiple Testing in Adaptive Trial Designs

Christopher Jennison

Department of Mathematical Sciences,

University of Bath, UK

<http://people.bath.ac.uk/mascj>

Takeda, London

20 March 2012

Outline of talk

1. Adaptive designs: Changing, or selecting, the null hypothesis
2. Closed testing procedures
3. Combination tests
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Examples of enrichment designs

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5. Seamless Phase IIb / Phase III designs

Methods using combination tests

Thall, Simon and Ellenberg's method

Assessing the benefits of seamless designs

1. Adaptive methods for a change of null hypothesis

There may be reasons to change the null hypothesis, or switch focus to a particular hypothesis out of several possibilities, during the course of a clinical trial:

Change of treatment definition, or selecting a dose,

Change of endpoint,

Restriction to a sub-group of the patient population,

Switching from a test of superiority to a test of non-inferiority.

When H_0 changes, a positive outcome requires the new null hypothesis to be rejected — with allowance for this hypothesis being data-generated.

Methods must protect the type I error rate when there are multiple hypotheses.

Selected references:

Bauer & Köhne (*Biometrics*, 1994), Bretz, Schmidli et al. (*Biometrical Journal*, 2006), Schmidli, Bretz et al. (*Biometrical Journal*, 2006).

2. Testing multiple hypotheses

Closed testing procedures

Suppose there are k null hypotheses, $H_i: \theta_i \leq 0$ for $i = 1, \dots, k$.

A procedure's **familywise error rate** under a set of values $(\theta_1, \dots, \theta_k)$ is

$$Pr\{\text{Reject } H_i \text{ for some } i \text{ with } \theta_i \leq 0\} = Pr\{\text{Reject any true } H_i\}.$$

The familywise error rate is controlled strongly at level α if this error rate is at most α for all possible combinations of θ_i values. Then

$$Pr\{\text{Reject any true } H_i\} \leq \alpha \quad \text{for all } (\theta_1, \dots, \theta_k).$$

Using such a procedure, the probability of choosing to focus on the parameter θ_{i^*} and then falsely claiming significance for null hypothesis H_{i^*} is at most α .

Closed testing procedures (Marcus et al, *Biometrika*, 1976) provide strong control by combining level α tests of each H_i and of intersections of these hypotheses.

Closed testing procedures

For each subset I of $\{1, \dots, k\}$, define the intersection hypothesis

$$H_I = \cap_{i \in I} H_i.$$

Construct a level α test of each intersection hypothesis H_I , i.e., a test which rejects H_I with probability at most α whenever all hypotheses specified in H_I are true.

Closed testing procedure

The simple hypothesis $H_j: \theta_j \leq 0$ is rejected overall if, and only if, H_I is rejected for every set I containing index j .

Proof of strong control of familywise error rate

Let $\tilde{I}$ be the set of indices of all true hypotheses H_i . For a familywise error to be committed, $H_{\tilde{I}}$ must be rejected.

Since $H_{\tilde{I}}$ is true, $Pr\{\text{Reject } H_{\tilde{I}}\} = \alpha$ and, thus, the probability of a familywise error is no greater than α .

Testing an intersection hypothesis

We have a choice of methods to test an intersection hypothesis $H_I = \cap_{i \in I} H_i$.

For each $i \in I$, let P_i denote the 1-sided P-value for testing H_i .

Suppose H_I is the intersection of m simple hypotheses and denote the P-values for the m hypotheses in increasing order by $P_{(k)}$, $k = 1, \dots, m$.

Bonferroni adjustment

The overall P-value for testing H_I is

$$P_I = m P_{(1)}.$$

Simes' method (Biometrika, 1986):

The overall P-value for H_I is

$$P_I = \min_{k=1, \dots, m} (m P_{(k)} / k).$$

Bonferroni and Simes' methods

The Bonferroni adjustment is simple, but conservative.

In the definition of Simes' P-value,

$$P_I = \min_{k=1,\dots,m} (m P_{(k)} / k),$$

the term for $k = 1$ is $mP_{(1)}$, i.e., the Bonferroni adjusted P-value.

Other low P-values can reduce the overall result, e.g., if $P_{(2)}$ is only a little higher than $P_{(1)}$ so $P_{(2)}/2 < P_{(1)}$, then this will reduce P_I .

The Simes method is valid — and slightly conservative — when the P_i are independent or positively dependent.

Such positive dependence arises in a comparison of m treatments with a common control or in tests for a treatment effect in overlapping sub-populations.

In special cases, such as m comparisons with a control, a precise treatment of the dependence between tests is possible — as we shall see in the next method.

Dunnett's method (JASA, 1955)

Suppose m treatments are compared with a control and, for $i = 1, \dots, m$, the null hypothesis H_i states that treatment i is no better than the control.

We are to test the intersection hypothesis $H_I = \cap_{i \in I} H_i$.

Suppose responses are normal with known variance and sample sizes on each treatment and the control are equal.

Denote the Z -statistic arising from the test of H_i by Z_i .

In the least favourable case under H_I , each treatment effect is zero, so we have $Z_i \sim N(0, 1)$, $i = 1, \dots, m$, and $\text{Cov}(Z_i, Z_{i'}) = 0.5$ for $i \neq i'$.

Let $Z^* = \max_{i \in I} Z_i$ and let z^* be the value attained by Z^* .

The P-value for testing H_I using Dunnett's test is

$$\Pr\{\max_{i \in I} Z_i > z^*\}$$

under the above multivariate normal joint distribution for $Z_i, i \in I$.

3. Combination tests

Suppose we run a clinical trial adaptively in two stages:

The design of stage 1 is set, then this part of the trial is conducted.

Results from stage 1 are analysed.

External information can be brought in if appropriate.

Stage 2 design is set, informed by stage 1 results and external information.

Stage 2 is conducted.

Results from stage 2 are analysed.

How can we test a null hypothesis, H_0 , protecting the type I error rate?

Combination tests

Before the trial commences, define the null hypothesis.

Suppose a hypothesis concerns the parameter θ and we are to test $H_0: \theta \leq 0$ against $\theta > 0$, with type I error probability α at $\theta = 0$.

Define one-sided P-values P_1 and P_2 from hypothesis tests of H_0 based on stage 1 and stage 2 data, respectively.

Under $\theta = 0$

$$P_1 \sim U(0, 1),$$

Conditionally on all stage 1 data and the stage 2 design, $P_2 \sim U(0, 1)$.

Hence, P_1 and P_2 are independent $U(0, 1)$ variates.

Bauer & Köhne's inverse χ^2 combination test

Reference: Bauer & Köhne (*Biometrics*, 1994).

Initial design

Stipulate that Bauer & Köhne's inverse χ^2 combination test will be used.

Define the null hypothesis, H_0 .

Design stage 1, fixing sample size and test statistic for this stage.

Stage 1

Observe the one-sided P-value, P_1 , based on stage 1 data.

Design stage 2 in the light of stage 1 data.

Stage 2

Observe the one-sided P-value, P_2 , based on stage 2 data alone.

Under H_0 , $P_1 \sim U(0, 1)$, $P_2 \sim U(0, 1)$ and P_2 is independent of P_1 .

Bauer & Köhne's inverse χ^2 combination test

Bauer & Köhne's rejects H_0 for low values of $P_1 P_2$.

If $P \sim U(0, 1)$, then

$$-\ln(P) \sim \text{Exp}(1) = \frac{1}{2} \chi_2^2.$$

Thus, under H_0 ,

$$-\ln(P_1 P_2) \sim \frac{1}{2} \chi_4^2.$$

Combining the two P-values in an overall test, we reject H_0 if

$$-\ln(P_1 P_2) > \frac{1}{2} \chi_{4, 1-\alpha}^2.$$

This χ^2 test was originally proposed for combining results of several studies by R. A. Fisher (1932) *Statistical Methods for Research Workers*.

The inverse normal combination test

Initial design

Stipulate the **inverse normal test** will be used to test the null hypothesis, H_0 .

For a two-stage design, specify weights w_1 and w_2 where $w_1^2 + w_2^2 = 1$.

Design stage 1, fixing sample size and test statistic for this stage.

Stage 1

Observe the one-sided P-value, P_1 , based on stage 1 data.

Compute $Z_1 = \Phi^{-1}(1 - P_1)$.

Design stage 2 in the light of stage 1 data.

Stage 2

Observe the one-sided P-value, P_2 , based on stage 2 data alone.

Compute $Z_2 = \Phi^{-1}(1 - P_2)$.

Under H_0 , $Z_1 \sim N(0, 1)$, $Z_2 \sim N(0, 1)$ and Z_1 and Z_2 are independent.

The inverse normal combination test

Overall test

Under H_0 , Z_1 and Z_2 are independent $N(0, 1)$, so

$$w_1 Z_1 + w_2 Z_2 \sim N(0, 1).$$

Hence, for an overall one-sided test, with type I error rate α , we reject H_0 if

$$w_1 Z_1 + w_2 Z_2 > \Phi^{-1}(1 - \alpha).$$

An attractive property

If the weights w_1 and w_2 are proportional to the square roots of the stage 1 and 2 sample sizes (more generally, the square roots of the information for θ), then

$$w_1 Z_1 + w_2 Z_2$$

is the standard Z statistic based on the full data set at the end of stage 2.

This is a function of the sufficient statistic for θ and the test based on it will have the usual efficiency properties.

Using combination tests in a closed testing procedure

Suppose we have null hypotheses $H_i, i = 1, \dots, k$, and wish to control the familywise type I error rate at level α using a closed testing procedure.

We need a level α test for each intersection hypothesis

$$H_I = \cap_{i \in I} H_i$$

— a simple hypotheses H_j is a special case where $I = \{j\}$.

Suppose an adaptive trial has two stages. Then, for each H_I , we shall apply a combination test across the stages.

Each stage provides a P-value for H_I and we combine these by a pre-specified method, e.g., “an inverse normal combination test with equal weights”.

The definition of the P-value from the second stage must be set before the second stage commences, but it may depend on first stage responses.

4. Enrichment designs

Consider a new treatment developed for a target population, but which is expected to be particularly effective in an identified sub-population.

An “enrichment design” aims to identify differential treatment effects in patient subgroups and adapt the trial’s focus to subgroups where benefit is greatest.

In a clinical trial with enrichment we

Start by comparing the new treatment against control in the full population.

Examine responses at an interim stage.

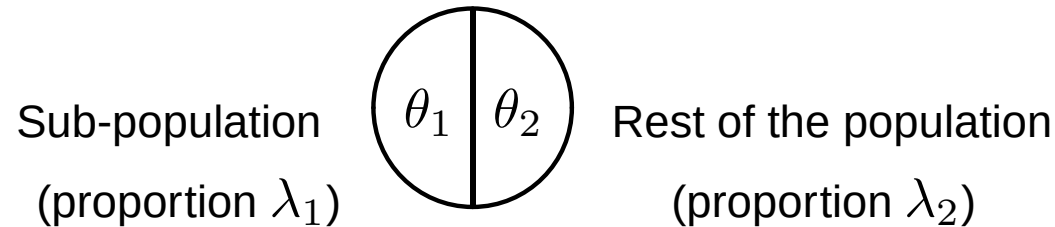
If there is no evidence of treatment effect, stop for futility.

If the new treatment appears effective in the full population, continue as before.

If the new treatment appears to benefit just the subgroup, recruit only from the subgroup and increase the numbers in this subgroup.

Results may support a licence for the full population or just the sub-population.

Enrichment designs



Denote the treatment effect:

In the sub-population by θ_1 ,

In the complement of the sub-population by θ_2 ,

Aggregated over the whole population by $\theta_3 = \lambda_1\theta_1 + \lambda_2\theta_2$.

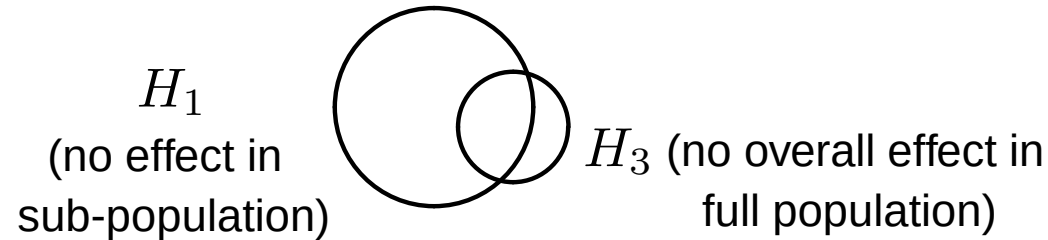
The null hypothesis for the sub-population is $H_1: \theta_1 \leq 0$.

The null hypothesis for the full target population is $H_3: \theta_3 \leq 0$.

The sets H_1 and H_3 are not necessarily nested.

All tests are against one-sided alternatives, $\theta_1 > 0$ or $\theta_3 > 0$.

Enrichment designs



Stage 1

P-value for H_1 from the sub-population data only is $P_{1,1}$.

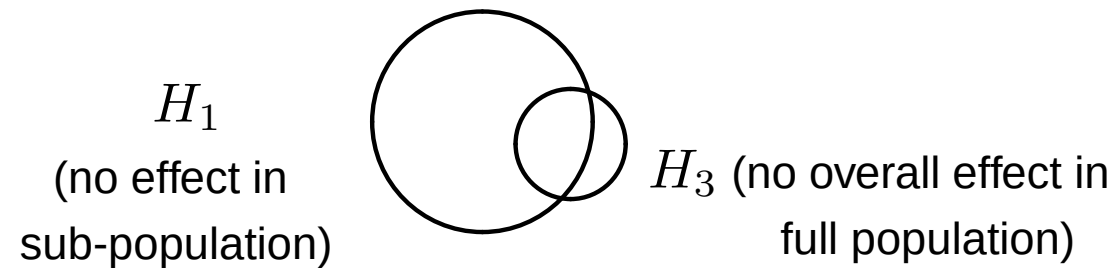
P-value for H_3 from the full data is $P_{1,3}$.

The P-value from Stage 1 data for $H_{13} = H_1 \cap H_3$ using Simes' method or, equivalently, Hochberg's adjustment (*Biometrika*, 1988) is

$$P_{1,13} = \min\{2 \min(P_{1,1}, P_{1,3}), \max(P_{1,1}, P_{1,3})\}.$$

This is conservative as $P_{1,1}$ and $P_{1,3}$ are positively correlated through their overlapping sets of subjects.

Enrichment designs



Case A: Stage 2 continuing with the full population

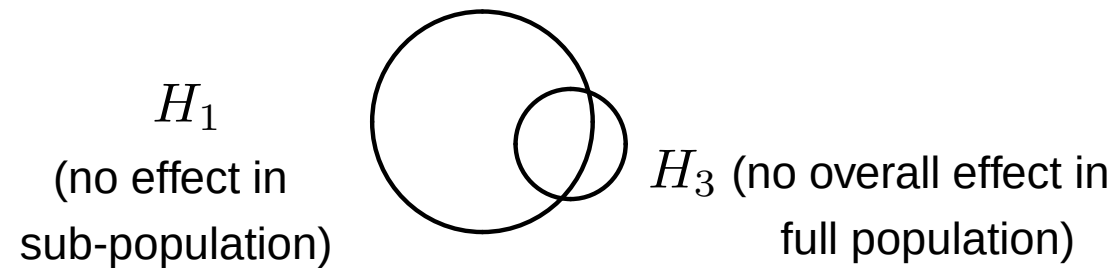
P-value for H_1 from the sub-population data only is $P_{2,1}$.

P-value for H_3 from the full Stage 2 data is $P_{2,3}$.

Simes' method give the conservative P-value for H_{13} from the Stage 2 data

$$P_{2,13} = \min\{2 \min(P_{2,1}, P_{2,3}), \max(P_{2,1}, P_{2,3})\}.$$

Enrichment designs



Case B: Stage 2 restricted to the sub-population

P-value for H_1 from the sub-population data is $P_{2,1}$.

No P-value is available for testing H_3 in Stage 2,

P-value for H_{13} from Stage 2 data is simply $P_{2,13} = P_{2,1}$.

Enrichment designs

Combining stages

A rule must be pre-specified for combining P-values from the two stages to give tests of H_1 , H_3 and H_{13} .

For example, with P-values p_1 and p_2 from Stages 1 and 2, a weighted inverse normal rule gives

$$Z(p_1, p_2) = w_1 \Phi^{-1}(1 - p_1) + w_2 \Phi^{-1}(1 - p_2),$$

The weights w_1 and w_2 reflect the relative sample sizes planned for Stage 1 and Stage 2 and satisfy $w_1^2 + w_2^2 = 1$.

Then $Z \sim N(0, 1)$ when P_1 and P_2 are independent $U(0, 1)$, and the relevant null hypothesis is rejected at level α if $Z > \Phi^{-1}(1 - \alpha)$.

Global tests of H_1 and H_3 with familywise error rate α are formed from individual level α tests of H_1 , H_3 and H_{13} , each combining P-values from the two stages.

Enrichment designs

Global tests

Case A: Stage 2 continuing with the full population

Global test of H_1 (sub-population effect)

H_1 is rejected at global level α if both H_1 and H_{13} are rejected (in combination tests for Stage 1 and Stage 2 data).

NB This is a stronger requirement than rejection of elementary hypothesis H_1 due to allowance for testing multiple hypotheses.

Global test of H_3 (full population effect)

H_3 is rejected at global level α if both H_3 and H_{13} are rejected.

As intended, it is possible to reject H_1 without rejecting H_3 and so find a treatment effect in the sub-population only.

Enrichment designs

Global tests

Case B: Stage 2 restricted to the sub-population

Global test of H_1 (sub-population effect)

H_1 is rejected at global level α if both H_1 and H_{13} are rejected.

No Stage 2 P-value is available for H_3 but this does not matter as a test of H_3 is not required in this case.

Enrichment designs: Example

We compare treatment vs control in a general population (null hypothesis H_3).

The treatment is expected to be particularly effective in men over 50 years old (null hypothesis H_1).

The familywise type I error rate is set as $\alpha = 0.025$.

Stage 1 and 2 data will be combined in an inverse normal test with $w_1 = w_2$.

Stage 1 results

Older men: $P_{1,1} = 0.02$

Full population: $P_{1,3} = 0.20$

Although the P-value for H_1 is low, it is one of two hypotheses under consideration.

Using Simes' method, the P-value for $H_1 \cap H_3$ from Stage 1 data is

$$P_{1,13} = \min\{2 \min(P_{1,1}, P_{1,3}), \max(P_{1,1}, P_{1,3})\} = 0.04.$$

We decide to continue to recruit from the full population in Stage 2.

Example, continued

Stage 2 results

$$\text{Older men: } P_{2,1} = 0.03$$

$$\text{Full population: } P_{2,3} = 0.10$$

$$P_{2,13} = \min\{2 \min(P_{2,1}, P_{2,3}), \max(P_{2,1}, P_{2,3})\} = 0.06.$$

Combining results from the two stages

$$Z_1 = \sqrt{0.5} \Phi^{-1}(1 - 0.02) + \sqrt{0.5} \Phi^{-1}(1 - 0.03) = 2.78, \quad P_1 = 0.003,$$

$$Z_3 = \sqrt{0.5} \Phi^{-1}(1 - 0.20) + \sqrt{0.5} \Phi^{-1}(1 - 0.10) = 1.50, \quad P_3 = 0.067,$$

$$Z_{13} = \sqrt{0.5} \Phi^{-1}(1 - 0.04) + \sqrt{0.5} \Phi^{-1}(1 - 0.06) = 2.34, \quad P_{13} = 0.010.$$

Both H_1 and H_{13} are rejected in individual tests at level $\alpha = 0.025$. Hence, H_1 is rejected in the global test controlling the familywise error probability at 0.025.

The adjusted P-value for H_1 is $\max\{P_1, P_{13}\} = 0.010$.

Elementary hypothesis H_3 has P-value $0.067 > \alpha$, so H_3 is not rejected overall.

Example, continued

Now suppose Stage 2 recruitment was restricted to the sub-population “Older men”.

Stage 1 results: As before

Stage 2 results

Older men: $P_{2,1} = 0.03$

Full population: $P_{2,3}$ is undefined

$$P_{2,13} = P_{2,1} = 0.03.$$

Combining results from the two stages

$$Z_1 = \sqrt{0.5} \Phi^{-1}(1 - 0.02) + \sqrt{0.5} \Phi^{-1}(1 - 0.03) = 2.78, \quad P_1 = 0.003,$$

$$Z_{13} = \sqrt{0.5} \Phi^{-1}(1 - 0.04) + \sqrt{0.5} \Phi^{-1}(1 - 0.03) = 2.57, \quad P_{13} = 0.005.$$

H_1 is rejected in the global test as both H_1 and H_{13} are rejected at $\alpha = 0.025$.

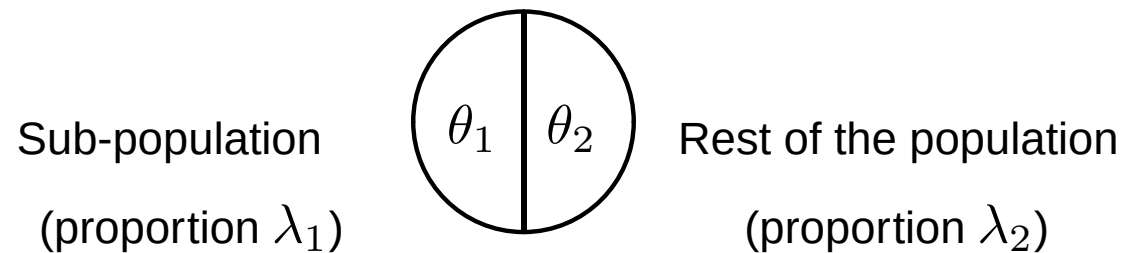
The adjusted P-value for H_1 is $\max\{P_1, P_{13}\} = 0.005$.

No combined test for H_3 is available since we have no value for $P_{2,3}$.

Assessing the benefits of enrichment designs

Consider the problem originally presented.

A trial is to investigate whether a new treatment is beneficial to the full population or, failing that, in a sub-population.



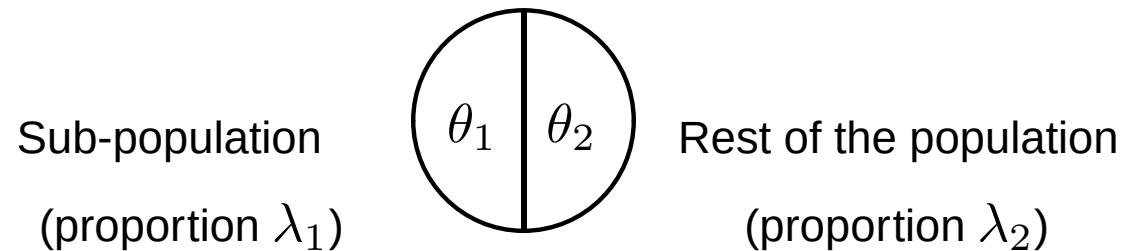
The treatment effect is θ_1 in the sub-population, θ_2 in its complement, and the average effect in the full population is $\theta_3 = \lambda_1\theta_1 + \lambda_2\theta_2$.

We wish to test:

The null hypothesis for the full population, $H_3: \theta_3 \leq 0$ vs $\theta_3 > 0$,

The null hypothesis for the sub-population, $H_1: \theta_1 \leq 0$ vs $\theta_1 > 0$.

The benefits of enrichment



First, consider a design testing for a whole population effect, $\theta_3 = \lambda_1 \theta_1 + \lambda_2 \theta_2$.

The design has two analyses and one-sided type I error probability 0.025.

Sample size is set to achieve power 0.9 at $\theta_3 = 20$.

Data in each stage are summarised by a Z -value:

	<i>Stage 1</i>	<i>Stage 2</i>	<i>Overall</i>
$H_3: \theta_3 \leq 0$	$Z_{1,3}$	$Z_{2,3}$	$Z_3 = \frac{1}{\sqrt{2}} Z_{1,3} + \frac{1}{\sqrt{2}} Z_{2,3}$

The benefits of enrichment

Two stage design testing for a whole population effect, θ_3 .

	<i>Stage 1</i>	<i>Stage 2</i>	<i>Overall</i>
$H_3: \theta_3 \leq 0$	$Z_{1,3}$	$Z_{2,3}$	$Z_3 = \frac{1}{\sqrt{2}}Z_{1,3} + \frac{1}{\sqrt{2}}Z_{2,3}$

Decision rules:

If $Z_{1,3} < 0$	Stop at Stage 1, Accept H_3
If $Z_{1,3} \geq 0$	Continue to Stage 2, then
If $Z_3 < 1.95$	Accept H_3
If $Z_3 \geq 1.95$	Reject H_3

The benefits of enrichment

Assume the sub-population comprises half the total population, so $\lambda_1 = \lambda_2 = 0.5$.

Properties of design for the whole population effect, θ_3 :

θ_1	θ_2	θ_3	Power for $H_3: \theta_3 \leq 0$
20	20	20	0.90
10	10	10	0.37
20	0	10	0.37

Is it feasible to identify at Stage 1 that θ_3 is low but θ_1 may be higher, so one might switch resources to test the sub-population?

The benefits of enrichment

We wish to be able to consider two null hypotheses:

$H_3: \theta_3 \leq 0$ Treatment is not effective in the whole population,

$H_1: \theta_1 \leq 0$ Treatment is not effective in the sub-population.

Since $\theta_3 = 0.5 \theta_1 + 0.5 \theta_2$, either of H_1 and H_3 may be true on its own.

In applying a ***closed testing procedure***, we also test the intersection hypothesis

$$H_{13}: \theta_1 \leq 0 \text{ and } \theta_3 \leq 0.$$

Then to reject H_1 overall, while protecting the family-wise type I error rate, we need to reject both H_1 and H_{13} in individual tests at significance level α .

Similarly, we can reject H_3 overall if both H_3 and H_{13} are rejected in level α tests.

An adaptive design

At Stage 1, if $\hat{\theta}_3 < 0$, stop to accept H_3 : $\theta_3 \leq 0$.

If $\hat{\theta}_3 > 0$ and the trial continues:

If $\hat{\theta}_2 < 0$ and $\hat{\theta}_1 > \hat{\theta}_2 + 8$ Restrict to sub-population 1 and test H_1 only,
needing to reject H_1 and H_{13} .

Else, Continue with full population and test H_3 ,
needing to reject H_3 and H_{13} .

The same *total* sample size for Stage 2 is retained in both cases, increasing the numbers for the sub-population when enrichment occurs.

An adaptive design

Each null hypothesis, H_i say, is tested in a 2-stage group sequential test.

With Z -statistics Z_1 and Z_2 from Stages 1 and 2, H_i is rejected if

$$Z_1 \geq 0 \quad \text{and} \quad \frac{1}{\sqrt{2}}Z_1 + \frac{1}{\sqrt{2}}Z_2 \geq 1.95.$$

When continuing with the full population, we use Z -statistics:

	Stage 1	Stage 2
H_3	$Z_{1,3}$	$Z_{2,3}$
H_{13}	$Z_{1,3}$	$Z_{2,3}$

where $Z_{i,3}$ is based on $\hat{\theta}_3$ from responses in Stage i .

With these definitions, there is no change from the original test of H_3 . This should help maintain power to reject H_3 and identify an effect in the full population.

An adaptive design

With Z -statistics Z_1 and Z_2 from Stages 1 and 2, H_i is rejected if

$$Z_1 \geq 0 \quad \text{and} \quad \frac{1}{\sqrt{2}}Z_1 + \frac{1}{\sqrt{2}}Z_2 \geq 1.95.$$

When switching to the sub-population, we use:

	Stage 1	Stage 2
H_1	$Z_{1,1}$	$Z_{2,1}$
H_{13}	$Z_{1,3}$	$Z_{2,1}$

where $Z_{i,j}$ is based on $\hat{\theta}_j$ from responses in Stage i .

The need to reject the intersection hypothesis H_{13} adds an extra requirement to the simple test of H_1 .

Simulation results: Power of non-adaptive and adaptive designs

	θ_1	θ_2	θ_3	<i>Non-adaptive</i>	<i>Adaptive</i>		
				<i>Full popⁿ</i>	<i>Sub-popⁿ only</i>	<i>Full popⁿ</i>	<i>Total</i>
1.	30	0	15	0.68	0.43	0.42	0.85
2.	20	0	10	0.37	0.24	0.26	0.51
3.	20	20	20	0.90	0.03	0.87	0.90
4.	20	10	15	0.68	0.11	0.60	0.71

Cases 1 & 2: Testing focuses (correctly) on H_1 , but it is still possible to find an effect (wrongly) for the full population. Overall power is increased.

Case 3: Restricting to the sub-population reduces power for finding an effect in the full population.

Case 4: Adaptation improves overall power a little.

Increasing power for finding a sub-population effect

Greater power to find an effect in the sub-population can be achieved by using $Z_{1,1}$ rather than $Z_{1,3}$ as the Stage 1 statistic in the test of H_{13} .

This gives the following results:

				<i>Non-adaptive</i>	<i>Adaptive</i>		
	θ_1	θ_2	θ_3	<i>Full popⁿ</i>	<i>Sub-popⁿ only</i>	<i>Full popⁿ</i>	<i>Total</i>
1.	30	0	15	0.68	0.47	0.40	0.87
2.	20	0	10	0.37	0.35	0.23	0.58
3.	20	20	20	0.90	0.04	0.74	0.78
4.	20	10	15	0.68	0.16	0.51	0.56

Benefits in Case 2 are balanced by loss of overall power in Cases 3 and 4.

Increasing power for finding a sub-population effect

As a compromise between the two previous methods, a combination* of $Z_{1,3}$ and $Z_{1,1}$ may be used as the Stage 1 statistic for the test of H_{13} .

This leads to the following results:

				<i>Non-adaptive</i>	<i>Adaptive</i>		
	θ_1	θ_2	θ_3	<i>Full popⁿ</i>	<i>Sub-popⁿ only</i>	<i>Full popⁿ</i>	<i>Total</i>
1.	30	0	15	0.68	0.47	0.41	0.88
2.	20	0	10	0.37	0.33	0.25	0.58
3.	20	20	20	0.90	0.04	0.83	0.87
4.	20	10	15	0.68	0.15	0.57	0.72

*Specifically, $(Z_{1,3} + Z_{1,1})/\sqrt{(2 + \sqrt{2})}$, which is $N(0, 1)$ under H_{13} .

The benefits of enrichment

In defining an enrichment design, the rules for staying with the full population or switching to the sub-population can be adjusted to favour specific goals.

However, we cannot eliminate the probability of making an error in these decisions.

This is to be expected. The standard error of the interim estimates $\hat{\theta}_1$ and $\hat{\theta}_2$ is 12.3 — much higher than the differences between θ_1 and θ_2 that interest us.

So, although restricting attention to a sub-population can be effective in improving power, higher overall sample size is needed for accurate sub-population inference.

Increasing sample size to achieve better overall performance

To match the non-adaptive test in cases 2 and 3, and obtain the benefits of adaptation elsewhere, increase the overall sample size by 30%.

With a combination* of $Z_{1,3}$ and $Z_{1,1}$ as the Stage 1 statistic for testing H_{13} , we obtain the following results:

				<i>Non-adaptive</i>	<i>Adaptive, 1.3 x sample size</i>		
	θ_1	θ_2	θ_3	<i>Full popⁿ</i>	<i>Sub-popⁿ only</i>	<i>Full popⁿ</i>	<i>Total</i>
1.	30	0	15	0.68	0.49	0.45	0.94
2.	20	0	10	0.37	0.38	0.30	0.69
3.	20	20	20	0.90	0.03	0.92	0.94
4.	20	10	15	0.68	0.15	0.68	0.82

*Using $(Z_{1,3} + Z_{1,1})/\sqrt{(2 + \sqrt{2})}$.

Conclusions: Enrichment designs

- With the advent of biomarkers and personalised medicine, there is a need for methods which allow adaptation to focus on a subgroup of patients during a clinical trial.
- Such methods are feasible using combination tests and a closed testing procedure.
- Restricting attention to a sub-population can be effective in improving power.
- However, higher overall sample size is needed for accurate sub-population inference.
- Open problems remain regarding more complex trial scenarios and inference on termination.

5. Seamless Phase IIb / Phase III designs

The roles of Phase IIb and Phase III trials

Phase IIb

Several dose levels of a treatment are compared against a control, to select a dose (or variant of a treatment) and confirm there is improvement against the control.

A Phase III

The trial is run as a confirmatory study to demonstrate superiority against control of the treatment selected in Phase IIb.

Stages:

Write Phase IIb protocol, seek ethical and regulatory approval, (FDA, IRBs, ...)

Run Phase IIb, analyse data, reach conclusions.

Write Phase III protocol, seek ethical and regulatory approval, (FDA, IRBs, ...)

Run Phase III, analyse data, reach final conclusion.

Planning the Phase III trial after Phase IIb is complete

Planning the Phase III trial after Phase IIb allows investigators to make use of information gained in Phase IIb.

They may decide to modify:

Treatment definition,

Target population,

Primary endpoint,

Sample size.

Positive results in Phase IIb will help recruitment for participation in Phase III.

But, planning and gaining approval for the Phase III trial can be time-consuming.

If the final outcome is positive, the sooner this conclusion is reached, the better.

Seamless Phase IIb / Phase III trials

Requirements

A single protocol for the combined Phase IIb and Phase III trials.

Rules for a committee managing the trials to follow as they:

Decide whether to proceed to Phase III,

Select the dose level for Phase III, considering efficacy and safety data,

Respond to information from Phase IIb, e.g., using estimated response variance to set Phase III sample size.

NB Expect everyone else to be blinded to the Phase IIb results.

Potential benefits

Eliminating the “white space” between phases,

Gaining efficiency from using Phase IIb data in the Phase III analysis.

Applying a closed testing procedure

Reference: Bretz, Schmidli et al. (*Biometrical Journal*, 2006).

Let θ_i , $i = 1, \dots, k$, denote the effect size of dose level i vs the control treatment.

Define H_i : $\theta_i \leq 0$ for $i = 1, \dots, k$.

Procedure

Select dose level i^* to advance to Phase III.

Test H_{i^*} : $\theta_{i^*} \leq 0$ at significance level α , with allowance for having selected H_{i^*} from k null hypotheses based on the results for dose i^* .

Formally

We shall define a procedure controlling the **familywise error rate**.

Then, for all possible sets of treatment effects $\{\theta_i\}$

$$Pr\{\text{Reject any true } H_i\} \leq \alpha.$$

Applying a closed testing procedure

Phase IIb

Observe estimated treatment effects $\hat{\theta}_{1,i}$, $i = 1, \dots, k$.

Select dose level i^* to go forward to Phase III.

Dose level i^* will have a high estimate $\hat{\theta}_{1,i^*}$ and good safety outcomes.

Phase III

Test dose level i^* against control.

In order to reject $H_{i^*}: \theta_{i^*} \leq 0$, we need to reject each intersection hypothesis H_I with $i^* \in I$ at level α .

Here, $H_I = \cap_{i \in I} H_i$ states that $\theta_i \leq 0$ for all $i \in I$.

Applying a closed testing procedure

Formally:

In order to reject $H_{i^*}: \theta_{i^*} \leq 0$, we need to reject each intersection hypothesis H_I with $i^* \in I$ at level α , based on combined Phase IIb and Phase III data.

Intuitively:

Dose i^* is chosen for the good results observed at this dose in Phase IIb.

We must adjust for this selection effect when adding the Phase IIb data on dose level i^* to the final analysis after Phase III.

Under a global null hypothesis of no treatment effect at any dose, the Phase IIb data on dose i^* should be viewed as *possibly the best results out of k ineffective doses*, rather than typical results at a single, pre-specified dose.

Testing an intersection hypothesis $H_I: \theta_i \leq 0$ for all $i \in I$

- a) We need to test an intersection hypothesis.
- b) We need to combine data from two stages, Phase IIb and Phase III.

Take problem (b) first

Denote the P-value for testing H_I in Phase IIb by $P_{1,I}$.

Denote the P-value for testing H_I in Phase III by $P_{2,I}$.

An inverse χ^2 combination test rejects H_I if

$$-\log(P_{1,I} P_{2,I}) > \frac{1}{2} \chi_{4, 1-\alpha}^2.$$

An inverse normal combination test rejects H_I if

$$w_1 \Phi^{-1}(1 - P_1) + w_1 \Phi^{-1}(1 - P_2) > \Phi^{-1}(1 - \alpha).$$

Testing an intersection hypothesis $H_I: \theta_i \leq 0$ for all $i \in I$

(a) Testing H_I is most complex in Phase IIb

For each $i \in I$, let $P_{1,i}$ denote the 1-sided P-value for testing $H_i: \theta_i \leq 0$.

If H_I is the intersection of m simple hypotheses, denote the P-values for the m hypotheses in increasing order by $P_{1,(j)}$, $j = 1, \dots, m$.

Bonferroni adjustment

The overall P-value for testing H_I is

$$P_{1,I} = P_{1,(1)}/m.$$

Simes' method (Biometrika, 1986):

The overall P-value for H_I is

$$P_{1,I} = \min_{j=1,\dots,m} (m P_{1,(j)}/j).$$

Remarks on Simes' method

Using Simes' method:

We are interested in testing intersection hypotheses H_I where I contains i^* , the index of the dose level selected for Phase III.

The P-value for testing H_I , where I has m elements, is

$$P_{1,I} = \min_{j=1,\dots,m} (m P_{1,(j)} / j).$$

If treatment i^* has the highest $\hat{\theta}_{1,i}$ and smallest P-value of all k treatments, we have $P_{1,(1)} = P_{1,i^*}$ in any set I containing i^* .

The term $mP_{1,(j)}/j$ with $j = 1$ is mP_{1,i^*} , which is also the Bonferroni adjusted version of P_{1,i^*} .

Simes' method “borrows strength” from other dose levels, allowing their P-values to reduce the overall result: if a second dose level performs well, $P_{1,(2)}/2$ may be smaller than P_{1,i^*} , reducing $P_{1,I}$.

Dunnett's method (JASA, 1955)

Consider $H_I = \cap_{i \in I} H_i$, the intersection of m simple hypotheses. Each H_i concerns the comparison of one dose level against the control.

Suppose responses are normal with known variance and sample sizes on each treatment and the control are equal.

Denote the Z -statistic arising from the test of H_i by Z_i .

In the least favourable case under H_I each treatment effect is zero, so we have $Z_i \sim N(0, 1)$, $i = 1, \dots, m$, and $\text{Cov}(Z_i, Z_{i'}) = 0.5$ for $i \neq i'$.

Let $Z^* = \max_{i \in I} Z_i$ and let z^* be the value attained by Z^* .

The P-value for testing H_I using Dunnett's test is

$$\Pr\{\max_{i \in I} Z_i > z^*\}$$

under the above multivariate normal joint distribution for $Z_i, i \in I$.

Testing an intersection hypothesis $H_I: \theta_i \leq 0$ for all $i \in I$

We return to problem (a) and testing of H_I in Phase III

In order to reject $H_{i^*}: \theta_{i^*} \leq 0$, we need to reject each H_I with $i^* \in I$.

Only dose level i^* is studied in Phase III, so a test of such an H_I using Phase III data is based on $\hat{\theta}_{2,i^*}$ — and there is just one such test.

Hence, all H_I of interest have a common P-value in Phase III of $P_{2,I} = P_{2,i^*}$.

With $P_{2,I} = P_{2,i^*}$, the inverse χ^2 combination test of H_I is a function of $\log(P_{1,I} P_{2,I})$, while the inverse normal combination test depends on

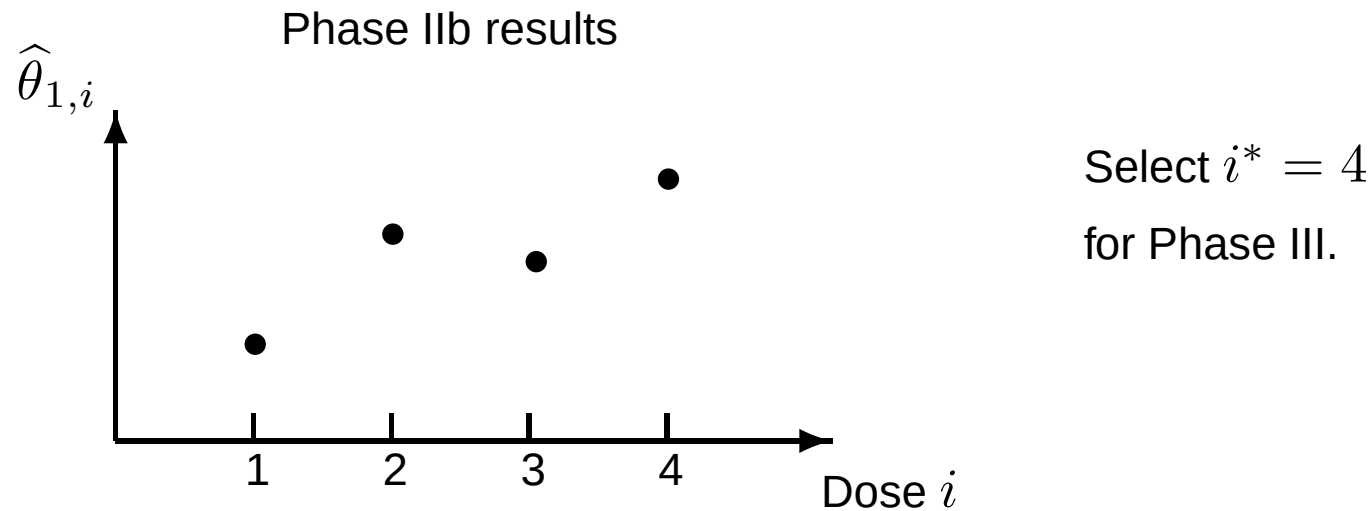
$$w_1 \Phi^{-1}(1 - P_{1,I}) + w_1 \Phi^{-1}(1 - P_{2,I}).$$

Rejection of H_{i^*} depends on the highest value of $P_{1,I}$ appearing in these formulae.

Thus, the key statistic from Phase IIb is:

$$\max_I P_{1,I} \text{ over sets } I \text{ containing } i^*.$$

Example of Simes' test



Suppose $P_{1,1} = 0.2$, $P_{1,2} = 0.04$, $P_{1,3} = 0.05$, $P_{1,4} = 0.03$.

For sets I containing $i^* = 4$, maximum $P_{1,I}$ comes from $I = \{1, 3, 4\}$.

Ordered P-values are $P_{1,(1)} = 0.03$, $P_{1,(2)} = 0.05$, $P_{1,(3)} = 0.2$:

$$P_{1,I} = \min_{j=1,\dots,3} (3 P_{1,(j)} / j) = 3 \times 0.05 / 2 = 0.075.$$

More details of Simes' test Example

Recall $P_{1,1} = 0.2$, $P_{1,2} = 0.04$, $P_{1,3} = 0.05$, $P_{1,4} = 0.03$.

Single element sets I containing $i^* = 4$

There is just one P-value $P_{1,4} = 0.03$, so $P_{1,I} = 0.03$.

Two-element sets I containing $i^* = 4$

Consider $I = \{1, 4\}$ with ordered P-values $P_{1,(1)} = 0.03$, $P_{1,(2)} = 0.2$:

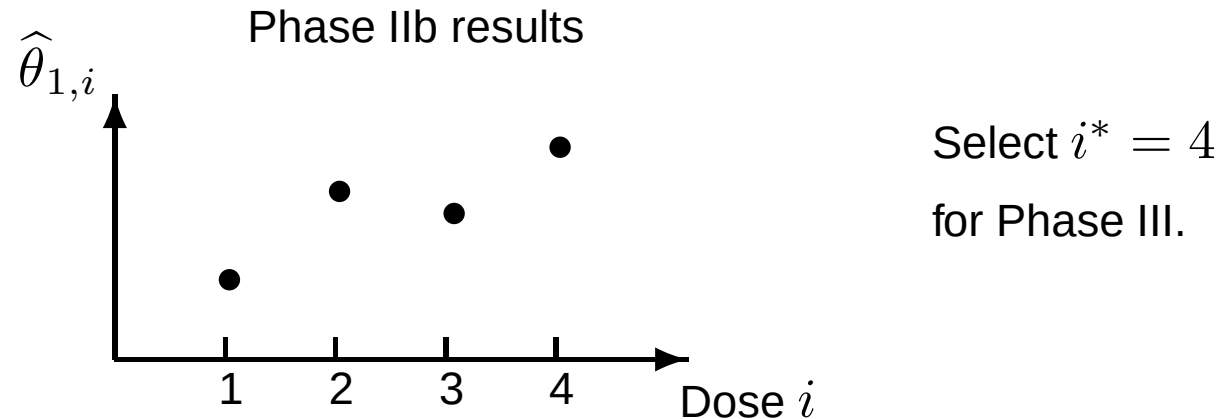
$$P_{1,I} = \min_{j=1,2} (2 P_{1,(j)} / j) = 2 \times 0.03 = 0.06.$$

Four-element sets I containing $i^* = 4$

Ordered P-values for $I = \{1, 2, 3, 4\}$ are $P_{1,(1)} = 0.03$, $P_{1,(2)} = 0.04$, $P_{1,(3)} = 0.05$, $P_{1,(4)} = 0.2$:

$$P_{1,I} = \min_{j=1,\dots,4} (4 P_{1,(j)} / j) = 4 \times 0.05 / 3 = 0.067.$$

Example of Dunnett's test



Suppose $P_{1,1} = 0.2$, $P_{1,2} = 0.04$, $P_{1,3} = 0.05$, $P_{1,4} = 0.03$.

For sets I containing $i^* = 4$, the maximum $P_{1,I}$ comes from $I = \{1, 2, 3, 4\}$.

The lowest P-value is $P_{1,4} = 0.03$, for which $\Phi^{-1}(1 - 0.03) = 1.881$.

The Dunnett test has P-value

$$P_{1,I} = Pr\left\{ \max_{i=1,\dots,4} (Z_i) > 1.881 \right\} = 0.0918,$$

calculated with $Z_i \sim N(0, 1)$, $i = 1, \dots, 4$, and $\text{Cov}(Z_i, Z_{i'}) = 0.5$ for $i \neq i'$.

Summary of combination of Phase IIb and Phase III data

In Phase IIb:

Select treatment i^* and carry forward $\max_{I:i^* \in I} P_{1,I}$.

In Phase III:

Test treatment i^* against control and find P_{2,i^*} .

Overall:

Combine the two P-values to see if $H_{i^*}: \theta_{i^*} \leq 0$ is rejected.

Flexibility:

Treatment i^* can be selected for efficacy, safety, or other factors

— not necessarily the treatment with maximum $\hat{\theta}_{1,i}$.

Efficiency:

Use of Phase IIb data in the final analysis should increase power

or reduce the Phase III sample size needed for a given power.

Other ways to combine Phase IIb and Phase III trials

Two-stage procedures for treatment selection and testing include proposals by:

Thall, Simon & Ellenberg (*Biometrika*, 1988),

Schaid, Wieand & Therneau (*Biometrika*, 1990), for survival endpoints,

Stallard & Todd (*Statistics in Medicine*, 2003), with group sequential testing in Phase III,

Sampson & Sill (*Biometrical Journal*, 2005).

These tests are not presented in terms of the closure principle, but they can be interpreted in that framework.

We shall focus first on the approach of Thall, Simon and Ellenberg.

See Jennison and Turnbull's (*Biometrical Journal*, 2006), discussion of the Bretz, Schmidli et al. papers for further details.

The method of Thall, Simon and Ellenberg (TSE)

Phase IIb

Take m_1 observations per treatment and control.

Denote estimated effect of treatment i against control by $\hat{\theta}_{1,i}$ and let the maximum of these be $\hat{\theta}_{1,i^*}$.

If $\hat{\theta}_{1,i^*} < C_1$, stop and accept $H_0: \theta_1 \leq 0, \dots, \theta_k \leq 0$,

if $\hat{\theta}_{1,i^*} \geq C_1$, select treatment i^* and proceed to Phase III.

Phase III

Take m_2 observations on treatment i^* and the control.

Combine data in $T_{i^*} = (m_1 \hat{\theta}_{1,i^*} + m_2 \hat{\theta}_{2,i^*}) / (m_1 + m_2)$.

If $T_{i^*} < C_2$, accept H_0 ,

if $T_{i^*} \geq C_2$, reject H_0 and conclude $\theta_{i^*} > 0$.

Thall, Simon and Ellenberg

NB The statistic T_{i^*} is the simple estimate of treatment θ_{i^*} obtained by pooling Phase IIb and Phase III data.

Type I error and power requirements imply values for m_1 , m_2 , C_1 and C_2 .

Type I error

Treatment i^* is said to be “chosen” if

Treatment i^* is selected at the end of Phase IIb, and

H_0 is rejected in favour of $\theta_{i^*} > 0$ in the final analysis.

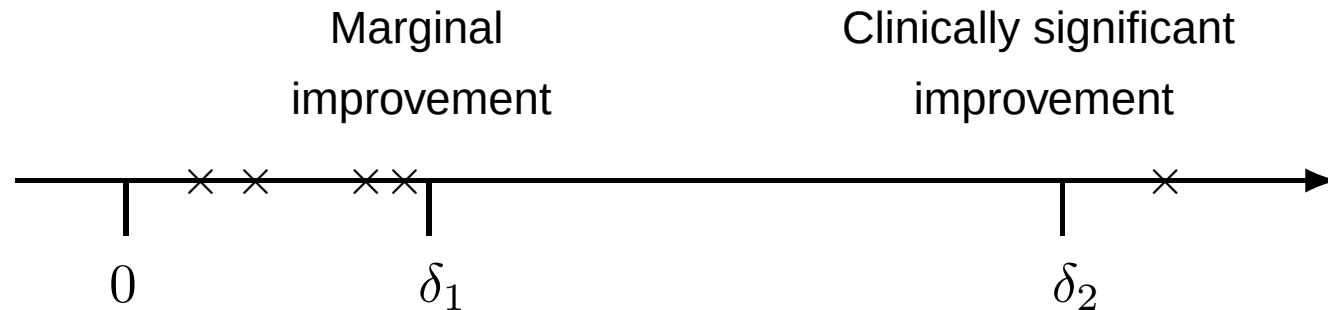
The type I error rate is the maximum value of

$$Pr\{\text{Any experimental treatment is “chosen”}\}$$

under $H_0: \theta_1 \leq 0, \dots, \theta_k \leq 0$, which occurs when $\theta_1 = \dots \theta_k = 0$.

Thall, Simon and Ellenberg

Power



Power depends on the full vector $\boldsymbol{\theta} = (\theta_1, \dots, \theta_k)$.

Any treatment with $\theta_i \geq \delta_2$ is said to be “acceptable”.

Consider cases of $\boldsymbol{\theta}$ where:

At least one treatment is acceptable,

No θ_i lies in the interval (δ_1, δ_2) .

The power function is $Pr_{\boldsymbol{\theta}} \{ \text{An acceptable choice is made} \}$.

Thall, Simon and Ellenberg

Power

TSE show power is minimized in cases described above under the *least favourable configuration*:

$$\theta_1 = \dots = \theta_{k-1} = \delta_1 \quad \text{and} \quad \theta_k = \delta_2.$$

They call this configuration θ^* and set their power condition at $\theta = \theta^*$.

Numerical integration under H_0 and θ^* enables parameters m_1, m_2, C_1 and C_2 to be found satisfying type I error and power conditions.

Optimisation

Tests minimizing expected sample size averaged over these two cases are found by searching feasible parameter combinations.

Thall, Simon and Ellenberg

The TSE method has familywise error rate α

TSE set their type I error probability α when $\theta_1 = \dots = \theta_k = 0$.

Family wise error rate concerns the probability of “choosing” a treatment with $\theta_i \leq 0$ under any vector θ , including cases where $\theta_j > 0$ for some indices j .

Starting from the case $\theta_1 = \dots = \theta_k = 0$, one can show that moving some effects to values $\theta_i < 0$ decreases the familywise error probability.

Then, increasing other values to $\theta_j > 0$ reduces the error rate as there are fewer treatments with $\theta_i \leq 0$ and the chance of selecting one of these is decreased.

We conclude that the TSE method protects the familywise error rate at level α .

Thall, Simon and Ellenberg

The TSE method follows the Closure Principle

Define a test of H_I for I containing i^* as follows:

If stopping at Phase IIb, accept H_I .

When proceeding to Phase III, define

$$\begin{aligned} T_I &= \frac{m_1 \max_{i \in I} \hat{\theta}_{1,i} + m_2 \hat{\theta}_{2,i^*}}{m_1 + m_2} \\ &= \frac{m_1 \hat{\theta}_{1,i^*} + m_2 \hat{\theta}_{2,i^*}}{m_1 + m_2} = T_{i^*} \end{aligned}$$

and reject H_I for $T_I > c_I$, accept H_I otherwise.

The value c_I for a level α test of H_I is highest when $I = \{1, \dots, k\} = I_k$, say.

Rejecting H_{I_k} implies rejecting every other H_I with $i^* \in I$.

Equating C_2 and c_{I_k} , we see that “choosing” i^* is equivalent to rejecting H_{i^*} according to the Closure Principle with these tests of intersection hypotheses H_I .

Thall, Simon and Ellenberg

Remarks on the TSE method

The single summary of Phase IIb data is $\hat{\theta}_{1,i^*}$. This is combined with Phase III data in the overall estimate of θ_{i^*} using the final sufficient statistic for θ_{i^*} .

The critical value for the final test statistic allows for selecting treatment i^* as the best performer in Phase IIb.

The null distribution taken for $\hat{\theta}_{1,i^*}$ is that of the maximum of n estimated effects when the true effects are all zero.

Adding flexibility:

The TSE method can be used, as defined, when a treatment i^* is selected with $\hat{\theta}_{1,i^*}$ less than the highest $\hat{\theta}_{1,i}$. The type I error rate is then met conservatively.

Alternatively, the above definitions of T_I , etc., can be extended to create a less conservative procedure via the Closure Principle.

Efficiency gains from combining Phase IIb and Phase III

Seamless Phase IIb / III designs carry a high organisational cost, so it is important that they should provide substantial benefits.

It is therefore of interest to compare:

Separate Phase IIb and Phase III trials

versus

Combined design with Phase IIb data used at the end of Phase III.

Since the vector of treatment effects is k -dimensional, there are many scenarios to consider. We shall return to the issue of efficiency later, after we have considered which methods for combining data are the most effective.

The design problem

Your clinical leader asks you to write the statistical design section of the protocol for a Phase IIb / III trial. Here are the ground rules:

- There will be (at most) two stages (Phase IIb and Phase III).
- In Stage 1 there will be 4 competing treatments and a control group. There are funds for 500 patients — 100 per group. The program may be ended as futile if no treatments look promising, otherwise, the best treatment will be chosen for comparison with the control in Stage 2.
- In Stage 2 there will be 1000 subjects — 500 on the selected treatment and 500 on control.
- At the end, a decision will be made either to accept the treatment selected for testing in Stage 2, or to reject it for insufficient improvement over the control.
- The overall probability of accepting an experimental treatment when all four are equal to the control must be no greater than $\alpha = 0.025$.

Design problem: Points to consider

Which of the following strategies would you recommend and in what circumstances?

1. Conventional

Separate Phase IIb and Phase III trials with final decision based on the Phase III data alone.

Critical value for the final Z -statistic is set at $c = 1.86$ which gives overall type I error rate 0.025 when $\theta_1 = \dots = \theta_4 = 0$.

This is an example of “buying back α ”. It is helpful here as it makes competing procedures properly comparable.

2. Bauer and Köhne

Combine results from Phases IIb and III using a Bauer and Köhne combination test and closed testing procedure with familywise type I error rate 0.025.

Design problem: Comparison of power functions

2. Bauer and Köhne, continued

For this method, it is necessary to specify:

Inverse χ^2 or inverse normal combination test,

Simes' test or Dunnett's test for intersection hypotheses.

In each case, the overall significance level applied should be adjusted to give a familywise type I error rate of 0.025.

3. Thall, Simon and Ellenberg

A TSE design with parameters $C_1 = 0$ and $C_2 = 0.449$, which give familywise type I error rate 0.025.

This is equivalent to a Stallard & Todd (*SiM*, 2003) design with two stages.

Design problem: Comparison of power functions

Definition of power

We define power to be the probability of selecting the best treatment and then rejecting its null hypothesis.

In comparing power of different designs, we first consider cases $\theta = (0, 0, 0, \delta)$ with $\delta > 0$, so only treatment 4 gives a benefit over the control.

Power can be calculated analytically for some designs. For example, the power of the Conventional design is

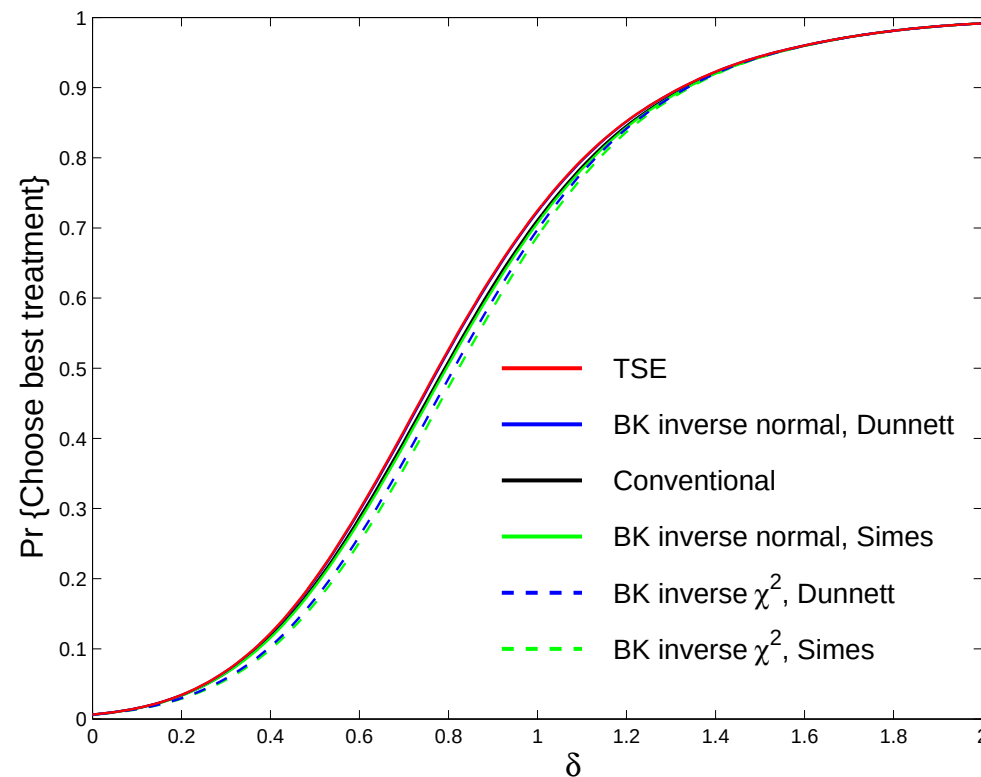
$$Pr\{\text{Treatment 4 picked at Stage 1}\} \times Pr\{\text{Treatment 4 picked at Stage 2}\}$$

and both terms can be computed fairly easily.

For combination tests, it is usually necessary to use simulation. When using simulation, we have run one million replicates so that standard errors of estimates of power are negligible.

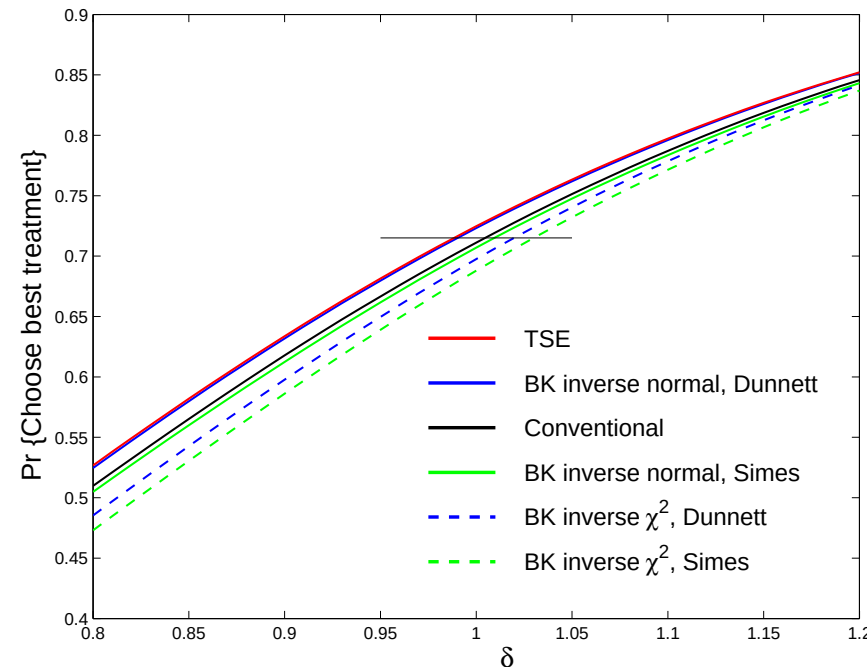
Power functions of the six selection and testing procedures

Power of six 2-stage selection/testing procedures when $\theta = (0, 0, 0, \delta)$, i.e., three treatments are ineffective and the other has effect size δ :



NB The sample size distribution is the same for all six methods.

Power functions of the six selection and testing procedures



The Thall, Simon & Ellenberg design has slightly higher power than the inverse normal combination test using a Dunnett rule.

The Conventional procedure — with no data combination — is only slightly worse than the TSE design, and superior to three versions of the BK combination test!

Differences in power equate to differences in sample size of around 4% to 8%.

Discussion of six power functions

- The variations in performance between procedures are surprising.
- While one BK method improves on the Conventional design, the others fail to gain any advantage from using Phase IIb data in the final hypothesis test.
Given the practical difficulties involved in a seamless Phase IIb / III design, a positive benefit is crucial.
- The TSE method and the BK design with weighted inverse normal combination test and Dunnett's test of intersection hypotheses perform well for mean vectors of the form $\theta = (0, 0, 0, \delta)$.

Are these methods also the best choices for other configurations of treatment means?

Is it possible to quantify the value of including Phase IIb data in the final analysis, e.g., in terms of an equivalent number of Phase III observations?

Assessing the benefits of seamless designs

Reference: Hampson & Jennison, *paper in preparation*.

It is possible to create a decision theoretic formulation of the testing problem at the end of a Phase IIb / Phase III trial.

Hampson & Jennison (HJ) have done so to find the decision rule which

Has type I error rate α under $\theta = (0, 0, 0, 0)$,

Maximises power under permutations of $\theta = (0, 0, 0, \delta)$ among all tests with type I error rate α .

With m_1 observations per treatment and control in Phase IIb, and m_2 observations on the selected treatment, i^* , and control in Phase III, the optimal procedure rejects H_{i^*} for high values of

$$\frac{8 m_1}{m_1 + m_2} \{ \hat{\theta}_{1,i^*} - \sum_{j \neq i^*} \frac{1}{4} \hat{\theta}_{1,j} \} + \frac{5 m_2}{m_1 + m_2} \hat{\theta}_{2,i^*}.$$

An “optimal” decision rule

The optimal procedure for the stated problem rejects H_{i^*} for high values of

$$\frac{8m_1}{m_1 + m_2} \{\hat{\theta}_{1,i^*} - \sum_{j \neq i^*} \frac{1}{4} \hat{\theta}_{1,j}\} + \frac{5m_2}{m_1 + m_2} \hat{\theta}_{2,i^*}.$$

Note the negative weights for terms $\hat{\theta}_{1,j}$, i.e., the estimated effects for treatments other than i^* .

This is due to their positive correlations with $\hat{\theta}_{1,i^*}$ through the common control.

A problem:

Although the type I error rate is protected at $\theta = (0, 0, 0, 0)$, this test will have higher type I error probability when some elements of θ are negative.

Thus, this test does not give the desired control of familywise type I error rate for all mean vectors θ .

An “optimal” decision rule

It is still instructive to consider the procedure that rejects H_{i^*} for high values of

$$\frac{8m_1}{m_1 + m_2} \{ \hat{\theta}_{1,i^*} - \sum_{j \neq i^*} \frac{1}{4} \hat{\theta}_{1,j} \} + \frac{5m_2}{m_1 + m_2} \hat{\theta}_{2,i^*}.$$

Combination tests using Simes' method to test intersection hypotheses try to “borrow strength” from treatments other than i^* — but the above rule shows that H_{i^*} should be rejected when other estimates $\hat{\theta}_{1,j}$ are *low*.

The TSE procedure gives zero weight to the other $\hat{\theta}_{1,j}$ s, and so is closer to the “optimal” decision rule.

The Conventional rule ignores all Phase IIb estimates, so at least it does not suffer from including terms $\hat{\theta}_{1,j}$ with the wrong sign.

Truly optimal decision rules

HJ's decision theory methodology can be applied to other forms of mean vector θ .

In certain cases, the optimal rule gives positive weight to all the $\hat{\theta}_{1,j}$ — then, this rule *does* protect the familywise type I error rate and so provides a benchmark for other methods.

Suppose θ is a permutation of $(\gamma \delta, \gamma \delta, \gamma \delta, \delta)$ for some $0 \leq \gamma < 1$.

If $\gamma \geq 0.5$, the optimal rule gives positive weight to all the $\hat{\theta}_{1,j}$.

In the case $\gamma = 0.5$, the optimal rule is exactly that of the TSE procedure.

We have also been able to derive optimal rules when θ is a permutation of $(\gamma_1, \gamma_2, \gamma_3, 1) \delta$, for some $0 \leq \gamma_1 \leq \gamma_2 \leq \gamma_3 < 1$.

In general, we have found the TSE procedure and the inverse normal combination test using a Dunnett rule to be highly efficient compared to optimal rules.

Relative efficiencies of 6 rules: $m_1 = 100, m_2 = 500$

Decision rule	Configuration of θ			
	A	B	C	D
TSE	100	100	99	99
BK inverse normal, Dunnett	100	100	100	99
BK inverse χ^2 , Dunnett	94	95	95	94
BK inverse normal, Simes	96	99	99	99
BK inverse χ^2 , Simes	92	94	94	94
Phase III data only	97	95	92	91

A: $\theta = (0, 0, 0, \delta)$

B: $\theta = (\gamma, \gamma, \gamma, 1) \delta, \gamma = 0.5$

C: $\theta = (\gamma, \gamma, \gamma, 1) \delta, \gamma = 0.75$

D: $\theta = (\gamma_1, \gamma_2, \gamma_3, 1) \delta, \gamma_1 = 0.75, \gamma_2 = 0.83, \gamma_3 = 0.92$

Relative efficiencies of 6 rules: $m_1 = 200, m_2 = 400$

Decision rule	Configuration of θ			
	A	B	C	D
TSE	100	100	99	98
BK inverse normal, Dunnett	99	100	99	98
BK inverse χ^2 , Dunnett	95	97	96	95
BK inverse normal, Simes	92	98	99	99
BK inverse χ^2 , Simes	93	96	96	95
Phase III data only	88	87	84	81

A: $\theta = (0, 0, 0, \delta)$

B: $\theta = (\gamma, \gamma, \gamma, 1) \delta, \gamma = 0.5$

C: $\theta = (\gamma, \gamma, \gamma, 1) \delta, \gamma = 0.75$

D: $\theta = (\gamma_1, \gamma_2, \gamma_3, 1) \delta, \gamma_1 = 0.75, \gamma_2 = 0.83, \gamma_3 = 0.92$

Conclusions about the benefits of seamless designs

We conclude that the TSE design is an efficient method for conducting a seamless Phase IIb / Phase III trial.

Similar efficiency is achieved by a Bauer and Köhne design using the inverse normal combination test and a Dunnett rule for testing intersection hypotheses.

Replacing the Dunnett rule by a Simes rule can lead to inefficiency when some treatments have low effect sizes.

The value of Phase IIb data: A rule of thumb

We compared power obtained by the above seamless designs with that of a Conventional design with a higher Phase III sample size. The Conventional design needs around $m_1/2$ additional Phase III observations on the each of the treatment and control arms to attain the power of the seamless designs.

Thus, the Phase IIb data on the selected treatment and control provide about half their face value.

Further topics for Phase IIb / Phase III trials

Other variations on the preceding methods are possible

More than one treatment may be carried forward to Phase III.

Sequential monitoring could result in early elimination of inferior treatments in Phase IIb, or an early Phase III decision, either positive or negative.

Dose response modelling

When the objective of Phase IIb is dose selection, it is only natural to model the response curve.

It is not straightforward to combine such modelling with hypothesis tests at individual doses — but see, Bretz, Pinheiro & Branson (*Biometrics*, 2005)

A DIA (formerly PhRMA) working group has been looking into the joint planning of Phase IIb and Phase III trials from an over-arching perspective.

Conclusions: Seamless Phase IIb / Phase III designs

- Seamless design can be an attractive proposition to accelerate the drug development process, or to increase efficiency by using Phase IIb data in the final Phase III analysis.
- Using combination tests with a closed testing procedure allows data to be combined with protection of the familywise type I error rate.
- The TSE method (Thall, Simon & Ellenberg, *Biometrika*, 1988) offers an alternative approach.
- We have presented examples and discussed questions of analysis and design.
- Choice of method is important to realise the benefits of data combination.
- We have found the TSE method to be an efficient choice.
- Using an inverse normal combination test with Dunnett's test for intersection hypotheses in the closed testing procedure also gives good performance.